

Risk factors for persistent decline in eGFR of Japanese patients with diabetes: Analysis using a large-scale diabetes registry J-DREAMS

Kota Yamada^{1,2} , Mitsuru Ohsugi^{1,3*} , Yuichiro Ito⁴ , Hiroki Uchida⁴ , Takumi Lee⁵ , Kohjiro Ueki^{1,6} 

¹Department of Diabetes, Endocrinology and Metabolism, National Center for Global Health and Medicine (Currently National Center for Global Health and Medicine, Japan Institute for Health Security), Center Hospital, Toyama, Shinjuku-ku, Tokyo, Japan, ²Department of Diabetes & Metabolic Diseases, Graduate School of Medicine, The University of Tokyo, Bunkyo-ku, Tokyo, Japan, ³Diabetes and Metabolism Information Center, Research Institute, National Center for Global Health and Medicine (Currently National Institute for Global Health and Medicine, Japan Institute for Health Security), Toyama, Shinjuku-ku, Tokyo, Japan, ⁴Medical Affairs, Astellas Pharma Inc., Chuo-Ku, Tokyo, Japan, ⁵Development, Astellas Pharma Inc., Chuo-Ku, Tokyo, Japan, and ⁶Diabetes Research Center, Research Institute, National Center for Global Health and Medicine (Currently National Institute for Global Health and Medicine, Japan Institute for Health Security), Toyama, Shinjuku-ku, Tokyo, Japan

Keywords

Cardio-renal anemia syndrome,
Diabetic kidney disease, Low serum
albumin

*Correspondence

Mitsuru Ohsugi
Tel.: +81-3-3202-7181
Fax: +81-3-3202-7364
E-mail address:
moosugi@hosp.ncgm.go.jp

J Diabetes Investig 2026; 17: 242–255

doi: [10.1111/jdi.70214](https://doi.org/10.1111/jdi.70214)

ABSTRACT

Aims/Introduction: This study aimed to identify clinical predictors associated with sustained decline in estimated glomerular filtration rate (eGFR) among Japanese individuals diagnosed with diabetes mellitus.

Materials and Methods: We conducted a retrospective analysis using J-DREAMS (Japan Diabetes compREhensive database project based on an Advanced Electronic Medical record System), a large-scale registry integrated with electronic medical records. The study population included adults (≥ 18 years) with diabetes who had at least one documented visit at a referral institution between December 1, 2015, and March 31, 2021, with concurrent measurements of serum creatinine and hemoglobin.

Results: A total of 12,416 and 11,157 patients were eligible for assessment of eGFR change at 1 and 2 years, respectively. Multivariable logistic regression identified several consistent risk factors for $\geq 30\%$ eGFR reduction at both time points: low serum albumin, elevated triglycerides, anemia as defined by JSDT, diabetic retinopathy, CKD Stage G4 or higher, albuminuria category A3, and chronic heart failure. Additional predictors at 1 year included age under 65 and elevated HbA1c, while smoking, hypertension, and diabetic neuropathy were significant only at 2 years.

Conclusion: Beyond retinopathy and advanced kidney disease, anemia, hypoalbuminemia, and heart failure independently contributed to persistent eGFR decline, supporting the clinical relevance of cardio-renal anemia syndrome in diabetic populations.

INTRODUCTION

The global prevalence of diabetes mellitus (DM) continues to rise, with projections estimating 643 million affected individuals by 2030 and 783 million by 2045¹. People with DM are frequently diagnosed with comorbid conditions/complications². Diabetic kidney disease (DKD) is one of the most common and serious complications of DM, representing the leading cause of end-stage kidney disease worldwide^{3,4}. Decline in estimated glomerular filtration rate (eGFR) has been recognized as an independent predictor of cardiovascular and renal outcomes in patients with type 2 diabetes^{5,6}. Several studies have

identified anemia^{7–10} and hypoalbuminemia^{7,11–12} as important contributors to DKD progression. Meta-analysis of 20 international cohorts by Jiang *et al.*¹³ established a risk prediction model for early DKD, highlighting albuminuria, HbA1c, and triglycerides as key predictors. Krolewski *et al.*¹⁴ emphasized the phenomenon of rapid renal decline in diabetic nephropathy and Vistisen *et al.*¹⁵ reported progressive eGFR decline even in patients without albuminuria. In Japan, similar findings have been reported. Hirano *et al.*¹⁶ identified lower hemoglobin levels as an independent risk factor for rapid eGFR decline in patients with preserved renal function and normoalbuminuria. Yoshida *et al.*¹⁷ found that older age, higher basal eGFR, higher albumin-to-creatinine ratio (ACR), and higher systolic blood

Received 4 July 2025; revised 26 October 2025; accepted 12 November 2025

pressure were significantly associated with early decliners. Ueda *et al.*⁷ also demonstrated that anemia, hypoalbuminemia, and elevated blood pressure contribute to renal function deterioration in Japanese diabetic populations. However, no large-scale study reporting comprehensive risk factors for progressive renal function decline in Japanese diabetic patients has been conducted.

In recent years, the use of eGFR slope as a surrogate marker for renal prognosis has gained traction in both research and clinical settings. However, in real-world clinical practice, there remains a lack of consensus on how to calculate eGFR slope—particularly regarding the optimal duration and frequency of eGFR measurements. This ambiguity can limit its practical utility in routine care. Therefore, in this study, we aimed to identify risk factors for renal function decline by using a more intuitive and clinically interpretable outcome: $a \geq 30\%$ reduction in eGFR at 1 and 2 years. This approach may offer clearer guidance for clinicians in identifying high-risk patients and implementing timely interventions.

To address these questions, we conducted a retrospective analysis using the J-DREAMS database (Japan Diabetes comprehensive database project based on an Advanced Electronic Medical Record System)¹⁸, which integrates standardized clinical information from diabetologists across Japan. This study aims to identify clinical predictors associated with sustained eGFR decline in a large Japanese diabetic cohort, and was conducted as part of ‘Prevalence and risk factors for anemia in patients with diabetes mellitus in Japan: a registry study using J-DREAMS’.

MATERIAL AND METHODS

Study settings

J-DREAMS, an extensive registry specifically developed for individuals with diabetes, was initiated by the NCGM in partnership with the Japan Diabetes Society (JDS)¹⁸. Through the implementation of a standardized diabetes management template (SDMT) embedded within the electronic medical record (EMR) infrastructure, data acquisition is uniformly conducted across all participating institutions. The SDMT captures information on diabetes status and associated comorbidities or complications, adhering to rigorously validated criteria. Upon entry of data into the SDMT, an automated process compiles essential demographic characteristics, prescribed treatments, and laboratory test results into the database. The diagnosis of DM is based on the guidelines provided by the JDS¹⁹. All medical conditions are documented using distinct codes that are directly linked to disease names, including dedicated identifiers for type 1, type 2, and secondary forms of diabetes, rather than relying on the International Classification of Diseases (ICD) coding system.

Patients

Individuals diagnosed with diabetes who were registered in the J-DREAMS system were deemed eligible for inclusion if they

fulfilled the following conditions (Figure 1): (1) Age of 18 years or older; (2) confirmed diagnosis of diabetes mellitus; (3) receiving care as outpatients; (4) availability of data recorded in J-DREAMS between December 1, 2015, and March 31, 2021, with at least one instance of simultaneous measurement of serum creatinine and hemoglobin within a 3-month window.

Exclusion parameters included: (1) Hospitalized patients; (2) individuals with eGFR below $5 \text{ mL/min/1.73 m}^2$; (3) those undergoing renal replacement modalities such as hemodialysis, peritoneal dialysis, or kidney transplantation; (4) subjects with hepatic dysfunction, including chronic hepatitis; (5) patients diagnosed with autoimmune disorders such as systemic lupus erythematosus, rheumatoid arthritis, Sjögren's syndrome, or celiac disease; (6) cases of anemia not attributable to renal causes, including hemolytic anemia, pancytopenia, hemorrhagic anemia, or presence of erythroblasts; (7) use of hypoxia-inducible factor-prolyl hydroxylase (HIF-PH) inhibitors.

From the cohort meeting these criteria, we extracted core demographic and biochemical variables, such as age, sex, HbA1c, eGFR, body mass index, serum albumin, LDL-C, HDL-C, and triglycerides. Chronic kidney disease (CKD) classified as stages G1–G2 and albuminuria category A1 was defined as absence of DKD, whereas all other combinations were categorized as DKD. Furthermore, documentation of ‘yes’, ‘no’, or ‘unknown’ was required for clinical history of neuropathy, retinopathy, coronary artery disease (CAD), cerebrovascular disease, peripheral artery disease (PAD), and chronic heart failure (CHF), based on clinical guidelines or physician assessment. For diabetic retinopathy, ophthalmologists provided the diagnosis, which was finalized by the attending physician using shared information. Hypertension was defined as systolic blood pressure $\geq 140 \text{ mmHg}$ and/or diastolic pressure $\geq 90 \text{ mmHg}$, or the use of antihypertensive agents. Dyslipidemia was defined as LDL-C $\geq 120 \text{ mg/dL}$ and/or administration of lipid-lowering therapy. Anemia was determined according to the Japanese Society for Dialysis Therapy (JSDT) criteria²⁰: hemoglobin $<13.5 \text{ g/dL}$ for males aged 19–59, $<12.0 \text{ g/dL}$ for males aged 60–69, and $<11.0 \text{ g/dL}$ for males aged 70 and above; $<11.5 \text{ g/dL}$ for females aged 19–59, and $<10.5 \text{ g/dL}$ for females aged 60 and older.

Study outcomes

While it would ideally be preferable to monitor progression to end-stage kidney disease (ESKD), the annual incidence was low, and the duration required to accumulate a sufficient number of such events may exceed the study period. Given that microalbuminuria in the initial phases of DKD does not follow a consistent linear trajectory²¹, we selected an alternative outcome measure: a reduction in eGFR of 30% or more within 1 or 2 years. From the J-DREAMS dataset, we identified individuals who had two eGFR values recorded at intervals exceeding 360 days, along with a urine albumin-to-creatinine ratio measured within 90 days before or after the baseline blood test. If multiple urine tests were available within that window, the earliest result was used. We then assessed whether a $\geq 30\%$ decline

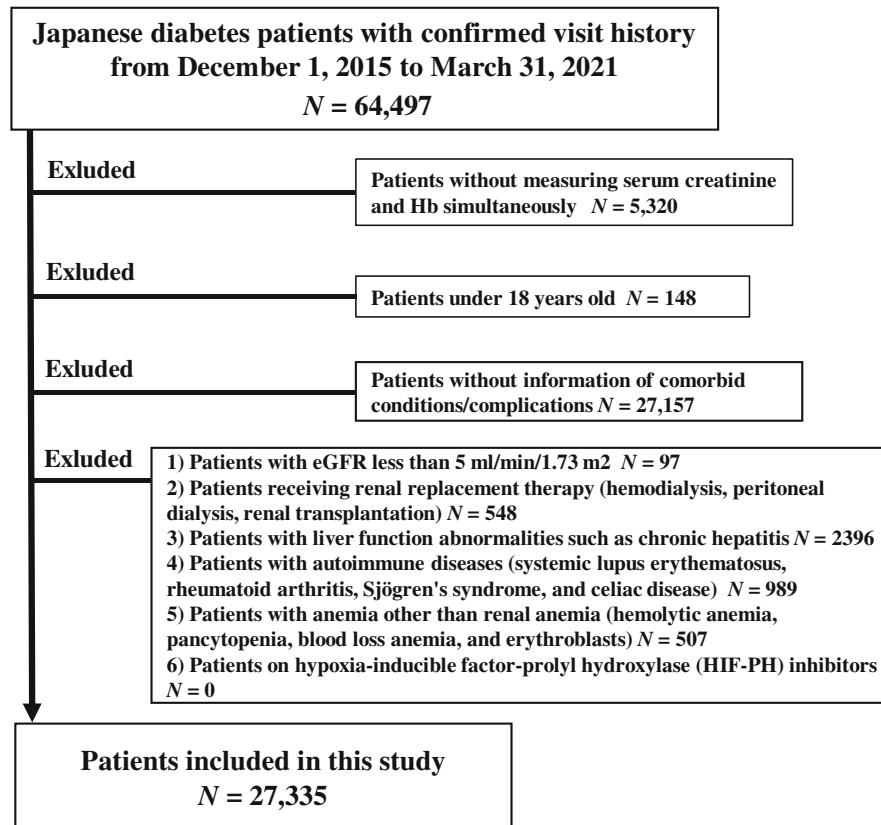


Figure 1 | Patient disposition 1.

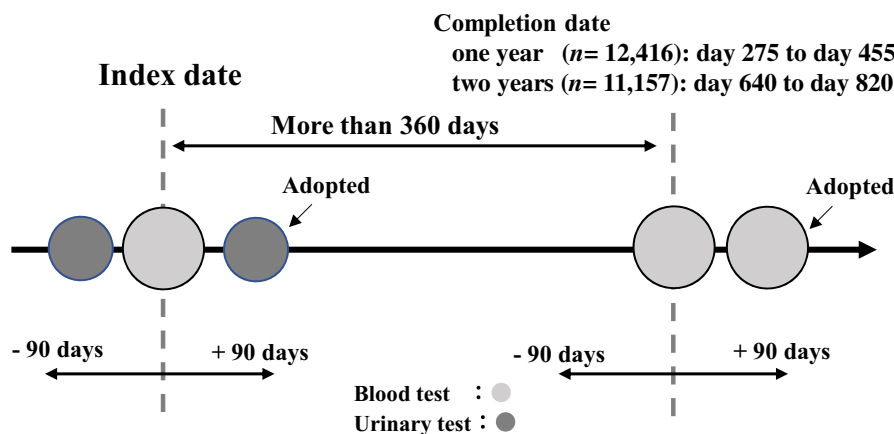


Figure 2 | Patient disposition 2.

in eGFR occurred at either 1 or 2 years. For the 1-year endpoint, patients were included if they had at least two eGFR values between 275 (365–90) and 455 (365 + 90) days from baseline, all showing a $\geq 30\%$ reduction. Similarly, for the 2-year endpoint, patients were included if they had values between 640 (730–90) and 820 (730 + 90) days from the index date.

Data analyses

To evaluate differences in patient profiles, comorbidities, and laboratory findings obtained from the J-DREAMS system, Welch's *t*-test was applied for continuous variables, while categorical variables were assessed using either the Chi-square test or Fisher's exact test, as appropriate. A *P*-value less than 0.05 was considered

Table 1 | Clinical and laboratory features, comorbid conditions/complications of the patients with or without $\geq 30\%$ decline in eGFR at 1 year

	Total patients eligible for eGFR change at 1 year (N = 12,416)		Patients with $>30\%$ decline in eGFR at 1 year (N = 353)		Patients without $>30\%$ decline in eGFR at 1 year (N = 12,063)		P-value
	n [†]	Value	n [†]	Value	n [†]	Value	
Age (years)	12,416	64.5 $\pm$ 13.8	353	62.2 $\pm$ 14.3	12,063	64.5 $\pm$ 13.7	0.002
Sex, female	12,416	4,882 (38.7)	353	112 (31.7)	12,063	4,690 (38.9)	0.007
Race, Japanese	12,416	12,416 (100)	353	353 (100)	12,063	12,063 (100)	
Duration of diabetes (years)	8,972	12.7 $\pm$ 10.66	268	11.3 $\pm$ 10.7	8,704	12.8 $\pm$ 10.6	<0.001
BMI (kg/m ²)	8,820	25.2 $\pm$ 4.9	236	24.8 $\pm$ 5.6	8,584	25.2 $\pm$ 4.9	0.2334
Smoking status							
Never	8,097	3,733 (46.1)	227	97 (42.7)	7,870	3,636 (46.2)	0.178
Current		1,700 (21.0)		59 (26.0)		1,641 (20.9)	
Past		2,664 (32.9)		71 (31.3)		2,593 (33.0)	
Drinking, nondrinkers	7,848	4,645 (59.2)	226	139 (61.5)	7,622	4,506 (59.1)	0.906
Type of diabetes							
Type 1	12,416	964 (7.8)	353	22 (6.2)	12,063	942 (7.8)	<0.001
Type 2		9,913 (74.0)		271 (76.8)		8,922 (74.0)	
Drug/other		771 (6.2)		30 (8.5)		741 (6.1)	
Unknown		1,488 (12.0)		30 (8.5)		1,458 (12.1)	
Systolic blood pressure(mmHg)	6,366	129.4 $\pm$ 16.2	195	132.1 $\pm$ 21.3	6,171	129.3 $\pm$ 16.0	0.0163
Diastolic blood pressure(mmHg)	6,346	74.0 $\pm$ 11.7	195	74.4 $\pm$ 12.8	6,151	74.0 $\pm$ 11.7	0.63
Fasting plasma glucose (mg/dL)	12,296	155.1 $\pm$ 66.0	349	189.8 $\pm$ 98.2	11,947	154.2 $\pm$ 64.5	<0.0001
HbA1c (%)	12,150	7.45 $\pm$ 1.5	346	8.28 $\pm$ 2.5	11,804	7.43 $\pm$ 1.4	<0.0001
Albumin (mg/dL)	9,090	4.1 $\pm$ 0.5	331	3.7 $\pm$ 0.6	8,759	4.1 $\pm$ 0.4	<0.0001
Total cholesterol (mg/dL)	10,594	188.5 $\pm$ 39.3	313	198.4 $\pm$ 55.6	10,281	188.2 $\pm$ 38.6	<0.0001
LDL-C (mg/dL)	9,187	104.3 $\pm$ 31.7	259	109.8 $\pm$ 43.5	8,928	104.1 $\pm$ 31.3	0.0043
Calculated LDL-C (mg/dL)	11,866	104.2 $\pm$ 31.6	324	111.0 $\pm$ 44.7	11,542	104.0 $\pm$ 31.1	0.0001
HDL-C (mg/dL)	11,736	56.8 $\pm$ 17.5	327	53.1 $\pm$ 17.3	11,409	56.9 $\pm$ 17.5	0.0001
TG (mg/dL)	11,963	144.5 $\pm$ 97.0	331	171.4 $\pm$ 123.4	11,632	143.8 $\pm$ 96.0	<0.0001
CPK (U/L)	20,016	112.3 $\pm$ 87.0	302	113.9 $\pm$ 104.1	8,771	115.6 $\pm$ 86.8	0.7419
AST (U/L)	12,065	25.9 $\pm$ 18.8	345	26.8 $\pm$ 33.2	11,720	25.9 $\pm$ 18.3	0.3853
ALT (U/L)	12,272	27.0 $\pm$ 24.7	346	27.4 $\pm$ 37.2	11,926	27.0 $\pm$ 24.3	0.7376
γ GTP (U/L)	11,619	44.7 $\pm$ 62.7	335	51.0 $\pm$ 73.0	11,284	44.5 $\pm$ 62.4	0.0608
Uric acid (mg/dL)	9,842	5.4 $\pm$ 1.4	281	5.6 $\pm$ 1.7	9,561	5.4 $\pm$ 1.4	0.0021
BUN (mg/dL)	11,125	17.0 $\pm$ 7.4	345	21.0 $\pm$ 12.9	10,780	16.9 $\pm$ 7.2	<0.0001
Creatinine (mg/dL)	12,416	0.88 $\pm$ 0.49	353	1.20 $\pm$ 0.93	12,063	0.87 $\pm$ 0.47	<0.0001
eGFR (mL/min/1.73 m ²)	12,401	70.5 $\pm$ 22.9	353	67.4 $\pm$ 34.9	12,048	70.5 $\pm$ 22.5	0.0118
K (mmol/L)	10,840	4.3 $\pm$ 0.4	342	4.3 $\pm$ 0.5	10,498	4.3 $\pm$ 0.4	0.0965
White blood cell (10 ³ / μ L)	12,416	6.8 $\pm$ 2.2	353	7.1 $\pm$ 2.5	12,063	6.8 $\pm$ 2.2	<0.0001
Hemoglobin (g/dL)	12,416	13.8 $\pm$ 1.8	353	12.8 $\pm$ 2.1	12,063	13.8 $\pm$ 1.8	<0.0001
Plateletes (10 ³ / μ L)	12,287	230.0 $\pm$ 64.5	348	230.4 $\pm$ 75.0	11,939	230.0 $\pm$ 64.1	0.9152
Hypertension	12,416	6,019 (48.5)	353	187 (53.0)	12,063	5,832 (48.3)	0.013
Dyslipidemia	12,416	6,045 (48.7)	353	165 (46.7)	12,063	5,880 (48.7)	0.106
JSDT anemia	12,416	1,122 (9.0)	353	109 (30.9)	12,063	1,013 (8.4)	<0.001
Diabetic neuropathy	12,416	1,995 (16.1)	353	91 (25.8)	12,063	1,904 (15.8)	<0.0001
Diabetic retinopathy	12,416	2,326 (18.7)	353	125 (35.4)	12,063	2,201 (18.2)	<0.0001
CKD G category							
G1 (≥ 90)	12,416	2,209 (17.8)	353	95 (26.9)	12,063	2,114 (17.5)	<0.001
G2 (60–89)		6,310 (50.8)		103 (29.2)		6,207 (51.5)	
G3a (45–59)		2,356 (19.0)		49 (13.9)		2,307 (19.1)	
G3b (30–44)		1,064 (8.6)		40 (11.3)		1,024 (8.5)	
G4 (15–30)		396 (3.2)		49 (13.9)		347 (2.9)	
G5 (<15)		81 (0.7)		17 (4.8)		64 (0.5)	

Table 1. (Continued)

	Total patients eligible for eGFR change at 1 year (N = 12,416)		Patients with >30% decline in eGFR at 1 year (N = 353)		Patients without >30% decline in eGFR at 1 year (N = 12,063)		P-value
	n [†]	Value	n [†]	Value	n [†]	Value	
CKD A category							
A1	10,203	7,067 (69.3)	329	100 (30.4)	9,924	6,967 (70.2)	<0.001
A2		1,085 (10.6)		77 (23.4)		1,058 (10.7)	
A3		2,051 (20.1)		152 (46.2)		1,899 (19.1)	
Coronary artery disease	12,416	1,460 (11.8)	353	44 (12.5)	12,063	1,416 (11.7)	0.634
Cerebrovascular disease	12,416	869 (7.0)	353	30 (8.5)	12,063	839 (7.0)	0.573
Peripheral artery disease	12,416	377 (3.0)	353	14 (4.0)	12,063	363 (3.0)	0.437
Chronic heart failure	12,416	493 (4.0)	353	37 (10.5)	12,063	456 (3.8)	<0.001

Values are expressed as the mean $\pm$ standard deviation or number (percent) of patients. [†]Number of patients with available data. A, albumin: creatinine ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; BUN, blood urea nitrogen; CKD, chronic kidney disease; CPK, creatine phosphokinase; eGFR, estimated glomerular filtration rate; G, grade; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein cholesterol; JSDT, Japanese Society for Dialysis Therapy; LDL-C, low-density lipoprotein cholesterol; TG, triglycerides; γ GTP, γ -glutamyl transpeptidase.

statistically significant. To identify predictors of $a \geq 30\%$ reduction in eGFR over 1 or 2 years, logistic regression models were employed to estimate adjusted odds ratios and corresponding 95% confidence intervals (CIs). Variables included in the models comprised patient demographics, clinical conditions, and laboratory parameters derived from the J-DREAMS dataset. Initially, multivariable logistic regression models were developed using variables that demonstrated a P -value ≤ 0.10 in univariable analyses. Subsequently, final models were constructed by selecting variables that retained a P -value ≤ 0.10 in the initial multivariable analysis. All statistical computations were performed using STATA/MP version 17.0 (Stata Corp., College Station, TX, USA).

Ethics

The study protocol underwent evaluation and endorsement by both the Institutional Review Board at the National Center for Global Health and Medicine (NCGM; approval number: NCGM-S-004378-00) and the Astellas Medical Affairs Japan Protocol Review Committee. This was a retrospective, noninterventional database study without patient involvement. Confidentiality was safeguarded by the NCGM. According to the Guidelines for Epidemiological Studies of the Ministry of Health, Labor and Welfare of Japan, written informed consent was not obligatory. Patients have access to the information about J-DREAMS on a website, and they have the right to withdraw their consent for data registration at any time.

RESULTS

Patients

Between December 1, 2015, and March 31, 2021, a total of 64,497 individuals with diabetes mellitus (DM) had documented visits recorded in the J-DREAMS system. Among them, 27,335 participants were included in the final study population (Figure 1). Of

these, 12,416 and 11,157 individuals met the criteria for evaluating changes in eGFR at the 2-year and 2-year time points, respectively (Figure 2). The number of patients who experienced a reduction in eGFR of 30% or more at 1 year was 353, while 547 patients met this threshold at the two-year mark.

Clinical and laboratory features, comorbid conditions/complications of the patients

Descriptive statistics for clinical profiles, laboratory indices, and associated health conditions of the study population are presented in Tables 1 and 2. At the 1- and 2-year follow-up intervals, the proportions of patients with diabetic neuropathy were 16.1% and 16.7%, respectively; diabetic retinopathy was observed in 18.7% and 19.8%; nephropathy Stages 2–4, as defined by the JDS, were present in 40.0% and 40.6%; CHF was noted in 4.0% and 4.1%; and anemia was found in 9.0% and 8.7%.

When comparing individuals who experienced $a \geq 30\%$ reduction in eGFR at 1 year with those who did not (Table 1), hypertension was more frequently observed in the former group, while the prevalence of dyslipidemia showed no significant difference. Diabetic neuropathy, retinopathy, and nephropathy Stage 3 or higher were more common among those with substantial eGFR decline. No significant differences were found in the rates of macrovascular complications such as coronary artery disease, cerebrovascular disease, or PAD, although CHF was more prevalent in the group with greater eGFR loss.

At the 2-year mark (Table 2), no notable differences in sex or average age were detected between patients with or without $\geq 30\%$ eGFR decline. However, those with greater decline had higher systolic blood pressure, longer diabetes duration, and a higher incidence of hypertension. Dyslipidemia rates remained similar between groups. Diabetic neuropathy, retinopathy, and advanced nephropathy were again more frequent in the group

Table 2 | Clinical and laboratory features, comorbid conditions/complications of the patients with or without $\geq 30\%$ decline in eGFR at 2 years

	Total patients eligible for eGFR change at 2 years (N = 11,157)		Patients with $>30\%$ decline in eGFR at 2 years (N = 547)		Patients without $>30\%$ decline in eGFR at 2 years (N = 10,610)		P-value
	n [†]	Value	n [†]	Value	n [†]	Value	
Age (years)	11,157	64.4 $\pm$ 13.5	547	64.0 $\pm$ 13.5	10,610	64.4 $\pm$ 13.5	0.4357
Sex, female	11,157	4,439 (39.8)	547	196 (35.8)	10,610	4,243 (40.0)	0.054
Race, Japanese	11,157	11,157 (100)	547	547 (100)	10,610	10,610 (100)	
Duration of diabetes (years)	8,485	13.4 $\pm$ 10.7	423	14.6 $\pm$ 11.9	8,062	13.3 $\pm$ 10.6	0.015
BMI (kg/m ²)	7,683	25.2 $\pm$ 4.9	382	24.9 $\pm$ 4.9	7,301	25.2 $\pm$ 4.9	0.3357
Smoking status							
Never	7,453	3,412 (45.8)	369	155 (42.0)	7,084	3,257 (46.0)	0.011
Current		1,531 (20.5)		99 (26.8)		1,432 (20.2)	
Past		2,510 (33.7)		115 (31.2)		2,395 (33.8)	
Drinking, nondrinkers	7,251	4,286 (59.1)	354	217 (61.3)	6,897	4,069 (59.0)	0.726
Type of diabetes							
Type 1	11,157	973 (8.7)	547	23 (4.2)	10,610	950 (9.0)	<0.001
Type 2		8,590 (77.0)		456 (83.4)		8,134 (76.7)	
Drug/other		619 (5.5)		31 (5.7)		588 (5.5)	
Unknown		975 (8.7)		37 (6.8)		938 (8.8)	
Systolic blood pressure (mmHg)	5,661	128.9 $\pm$ 16.0	295	133.9 $\pm$ 19.4	5,366	128.7 $\pm$ 15.8	<0.0001
Diastolic blood pressure (mmHg)	5,647	73.6 $\pm$ 11.6	295	73.2 $\pm$ 12.6	5,352	73.6 $\pm$ 11.6	0.5586
Fasting plasma glucose (mg/dL)	11,053	152.8 $\pm$ 61.8	543	177.2 $\pm$ 86.7	10,510	151.5 $\pm$ 60.0	<0.0001
HbA1c (%)	10,953	7.39 $\pm$ 1.4	535	7.78 $\pm$ 2.0	10,418	7.36 $\pm$ 1.3	<0.0001
Albumin (mg/dL)	8,117	4.1 $\pm$ 0.5	481	3.7 $\pm$ 0.6	7,636	4.2 $\pm$ 0.4	<0.0001
Total cholesterol (mg/dL)	9,650	186.9 $\pm$ 37.7	482	188.4 $\pm$ 49.5	9,168	186.8 $\pm$ 36.9	0.3504
LDL-C (mg/dL)	8,155	103.0 $\pm$ 30.4	421	102.1 $\pm$ 38.6	7,734	103.0 $\pm$ 29.9	0.5602
Calculated LDL-C (mg/dL)	10,792	103.3 $\pm$ 30.6	517	102.3 $\pm$ 38.9	10,275	103.4 $\pm$ 30.1	0.4338
HDL-C (mg/dL)	10,693	57.0 $\pm$ 17.6	515	54.3 $\pm$ 19.3	10,178	57.2 $\pm$ 17.5	0.0003
TG (mg/dL)	10,857	142.8 $\pm$ 95.5	525	173.2 $\pm$ 125.3	10,332	141.3 $\pm$ 93.5	<0.0001
CPK (U/L)	8,111	116.7 $\pm$ 85.6	420	119.8 $\pm$ 106.1	7,691	116.6 $\pm$ 84.3	0.4595
AST (U/L)	10,761	25.4 $\pm$ 17.0	539	24.5 $\pm$ 19.4	10,222	25.4 $\pm$ 16.9	0.234
ALT (U/L)	11,033	26.2 $\pm$ 22.6	542	24.6 $\pm$ 28.7	10,491	26.3 $\pm$ 22.2	0.0959
γ -GTP (U/L)	10,397	42.3 $\pm$ 59.7	519	47.7 $\pm$ 71.7	9,878	42.0 $\pm$ 59.0	0.0331
Uric acid (mg/dL)	8,515	5.4 $\pm$ 1.4	435	5.7 $\pm$ 1.7	8,080	5.4 $\pm$ 1.4	<0.0001
BUN (mg/dL)	9,901	16.9 $\pm$ 7.1	525	22.1 $\pm$ 12.1	9,376	16.6 $\pm$ 6.6	<0.0001
Creatinine (mg/dL)	11,157	0.87 $\pm$ 0.45	547	1.23 $\pm$ 0.86	10,610	0.85 $\pm$ 0.41	<0.0001
eGFR (mL/min/1.73 m ²)	11,157	70.3 $\pm$ 22.4	547	61.8 $\pm$ 33.1	10,610	70.7 $\pm$ 21.6	<0.0001
K (mmol/L)	9,754	4.35 $\pm$ 0.4	526	4.4 $\pm$ 0.5	9,228	4.35 $\pm$ 0.4	0.0125
White blood cell (10 ³ / μ L)	11,143	6.7 $\pm$ 2.1	547	7.2 $\pm$ 2.5	10,596	6.7 $\pm$ 2.1	0.0001
Hemoglobin (g/dL)	11,157	13.8 $\pm$ 1.7	547	12.6 $\pm$ 2.0	10,610	13.8 $\pm$ 1.7	<0.0001
Plateletes (10 ³ / μ L)	11,052	229.6 $\pm$ 63.4	536	228.0 $\pm$ 67.5	10,516	229.7 $\pm$ 63.2	0.5536
Hypertension	11,157	5,777 (51.7)	547	370 (67.6)	10,610	5,407 (51.0)	<0.001
Dyslipidemia	11,157	5,926 (53.1)	547	306 (55.9)	10,610	5,880 (55.4)	0.276
JSDT anemia	11,157	970 (8.7)	547	162 (29.6)	10,610	808 (7.6)	<0.001
Diabetic neuropathy	11,157	1,868 (16.7)	547	157 (28.7)	10,610	1,711 (16.1)	<0.001
Diabetic retinopathy	11,157	2,210 (19.8)	547	212 (38.8)	10,610	1,998 (18.8)	<0.001
CKD G category							
G1 (≥ 90)	11,157	1,912 (17.1)	547	118 (21.5)	10,610	1,794 (16.9)	<0.001
G2 (60–89)		5,737 (51.4)		150 (27.4)		5,587 (52.7)	
G3a (45–59)		2,153 (19.3)		77 (14.1)		2,076 (19.6)	
G3b (30–44)		963 (8.6)		88 (16.1)		875 (8.3)	
G4 (15–30)		345 (3.1)		95 (17.4)		250 (2.4)	
G5 (<15)		47 (0.4)		19 (3.5)		28 (0.3)	

Table 2. (Continued)

	Total patients eligible for eGFR change at 2 years (N = 11,157)		Patients with >30% decline in eGFR at 2 years (N = 547)		Patients without >30% decline in eGFR at 2 years (N = 10,610)		P-value
	n [†]	Value	n [†]	Value	n [†]	Value	
CKD A category							
A1	9,356	6,451 (69.0)	439	134 (30.5)	8,917	6,317 (70.8)	<0.001
A2		1,071 (11.5)		37 (8.4)		1,034 (11.6)	
A3		1,834 (19.6)		268 (61.1)		1,566 (17.6)	
Coronary artery disease	11,157	1,426 (12.8)	547	109 (19.9)	10,610	1,317 (12.4)	<0.001
Cerebrovascular disease	11,157	821 (7.4)	547	55 (10.1)	10,610	766 (7.2)	0.043
Peripheral artery disease	11,157	364 (3.3)	547	42 (7.7)	10,610	322 (3.0)	<0.001
Chronic heart failure	11,157	461 (4.1)	547	69 (12.6)	10,610	392 (3.7)	<0.001

Values are expressed as the mean $\pm$ standard deviation or number (percent) of patients. [†]Number of patients with available data. A, albumin: creatinine ratio; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; BUN, blood urea nitrogen; CKD, chronic kidney disease; CPK, creatine phosphokinase; eGFR, estimated glomerular filtration rate; G, grade; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein cholesterol; JSDT, Japanese Society for Dialysis Therapy; LDL-C, low-density lipoprotein cholesterol; TG, triglycerides; γ GTP, γ -glutamyl transpeptidase.

with significant eGFR reduction. Additionally, all macrovascular conditions—including coronary, cerebrovascular, and PAD, as well as CHF—were more common in patients with $\geq 30\%$ eGFR decline at 2 years.

Risk factors for eGFR decline $>30\%$ during 1 year or 2 years follow-up

To identify determinants of a $\geq 30\%$ reduction in eGFR over 1- and 2-year intervals, we conducted univariable logistic regression analyses using the presence or absence of such decline as the dependent variable. Explanatory variables included demographic characteristics (age, sex, diabetes duration, smoking status), laboratory parameters (HbA1c, serum albumin, HDL-C, triglycerides), CKD stage (G and A categories), and clinical comorbidities (hypertension, dyslipidemia, anemia, microvascular and macrovascular complications, and CHF), as detailed in Tables 3 and 4.

Variables with a P -value ≤ 0.10 in univariable analysis and deemed clinically relevant were entered into initial multivariable logistic regression models. Final models were then refined by retaining only those variables that remained significant at $P \leq 0.10$ in the first multivariable step.

In the final model for the 1-year cohort ($n = 5,294$), significant predictors of $\geq 30\%$ eGFR decline included age < 65 years, reduced serum albumin, elevated triglycerides, higher HbA1c, anemia per JSDT criteria, diabetic retinopathy, CKD stage G4 or worse, albuminuria category A3, and CHF. For the 2-year cohort ($n = 4,355$), risk factors included current smoking, low serum albumin, high triglycerides, hypertension, JSDT-defined anemia, diabetic neuropathy, retinopathy, CKD G4 or higher, A3 albuminuria, and CHF.

Shared predictors across both timeframes were hypoalbuminemia, hypertriglyceridemia, anemia, retinopathy, advanced CKD (G4 or higher and/or A3), and CHF. Younger age and

poor glycemic control were specific to the 1-year model, while smoking, hypertension, and neuropathy were unique to the 2-year model.

DISCUSSION

This investigation, leveraging the J-DREAMS dataset, tracked longitudinal changes in eGFR over 1- and 2-year periods and identified clinical factors associated with a sustained decline of $\geq 30\%$. Our findings highlight the consistent roles of anemia, hypoalbuminemia, elevated triglycerides, diabetic retinopathy, advanced CKD, and CHF in predicting renal deterioration. These results reinforce the concept of the cardio-renal anemia syndrome and its relevance in diabetic populations^{22,23}.

While eGFR slope has emerged as a valuable tool for assessing renal prognosis in research settings, its application in routine clinical practice remains limited. One major challenge is the lack of consensus regarding its calculation—particularly the duration and frequency of eGFR measurements to be used^{24,25}. Although the eGFR slope is a recommended metric for predicting the progression of CKD²⁶, its calculation often requires complex statistical modeling, such as mixed-effects models²⁷. This complexity can hinder its practical application in clinical settings. To address this, we adopted a more intuitive and time-bound outcome: a $\geq 30\%$ reduction in eGFR at 1 and 2 years. In contrast to complex slope calculations, our study utilized a more intuitive and accessible method for evaluating renal function decline. This approach offers clearer clinical relevance and may facilitate earlier identification of high-risk individuals.

Our findings are consistent with prior studies conducted in Japan, which have reported associations between anemia, hypoalbuminemia, and renal decline in diabetic patients⁷. However, most of these studies were limited by smaller sample sizes or focused on specific subgroups, such as patients with preserved renal function or normoalbuminuria. By leveraging the

Table 3 | Logistic regression analysis for identifying risk factors for $\geq 30\%$ decline in eGFR during 1 year follow-up

Univariable	Multivariable				
	Total no. in category	No. of events (%)	OR [95% CI]	P-value	n = 4012
Age	5,388	190	3.53	1.00	Age
	7,028	163	2.32	0.65 [0.53–0.80]	<65
Sex	7,614	241	3.17	1.00	65≤
	4,802	112	2.33	0.73 [0.58–0.92]	Men
Duration of diabetes	8,972			0.027	Women
				0.033	By 5 years
BMI	17	1	5.88	2.50 [0.33–19.06]	Duration of diabetes
	932	40	4.29	1.80 [1.23–1.11]	BMI
	3,939	96	2.44	1	<15 kg/m ²
	2,754	64	2.32	0.95 [0.69–1.31]	15–199 kg/m ²
	809	22	2.72	1.12 [0.70–1.79]	20–249 kg/m ²
	256	11	4.30	1.80 [0.95–3.40]	25–299 kg/m ²
	113	2	1.77	0.72 [0.18–2.96]	30–349 kg/m ²
Smoking	1,700	59	3.47	1.35 [0.97–1.87]	35–399 kg/m ²
	2,664	71	2.67	1.03 [0.75–1.40]	40 kg/m ² ≤
	3,733	97	2.60	1.00	Current
	12,150			1.29 [1.22–1.35]	Past
Alc	9,090			5.37 [4.44–6.48]	Never
Albumin	11,963			1.05 [1.03–1.07]	By 1.0%
Triglycerides	11,736			1.15 [1.07–1.23]	By –10 mg/dL
HDL-C	6,397			1.00	By 50 mg/dL
Hypertension	6,019			1.00	By –10 mg/dL
	6,371			1.00	Present
	6,045			1.00	Absent
	11,294			1.00	Present
	1,122			1.00	Absent
JSDT anemia	10,421			1.00	Present
	1,995			1.00	Absent
Diabetic neuropathy	2,326			1.00	Present
	2,209			1.00	Absent
Diabetic retinopathy	6,310			1.00	Present
	2,307			1.00	Absent
CKD G category	1,064			1.00	Present
	396			1.00	Absent
	81			1.00	Present
CKD A category	7,067			1.00	Absent
	1,085			1.00	Present
	2,051			1.00	Absent
CAD	10,956			1.00	Present
	1,460			1.00	Absent

Table 3. (Continued)

Univariable	Multivariable				AOR [95% CI]	P-value	n = 5,294	AOR [95% CI]	P-value
	Total no. in category	No. of events (%)	OR [95% CI]	P-value					
Cerebrovascular disease	Absent	11,547							
	Present	869	323 (37.1)	2.80	1.00				
PAD	Absent	12,039							
	Present	377	14 (3.7)	1.24 [0.85–1.82]	0.263				
CHF	Absent	11,923							
	Present	493	316 (63.3)	1.00	1.33 [0.77–2.29]	0.303			
					2.98 [2.09–4.24]	<0.001			

On the left side, the unadjusted odds ratio was calculated for each complication/comorbidity as the explanatory variable. Multivariable logistic regression analysis, employing each explanatory variable based on a significance level of $P \leq 0.1$, was performed to calculate the respective adjusted odds ratio, which is shown in the table on the middle side. Multiple logistic analysis was then performed again by selecting the variables with $P \leq 0.1$ in the middle analysis as explanatory variables, which are shown in the table on the right side. A, albumin; creatinine ratio; BMI, body mass index; CAD, coronary artery disease; CHF, chronic heart failure; CKD, chronic kidney disease; G, grade; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein cholesterol; PAD, peripheral artery disease; TG, triglycerides.

J-DREAMS database¹⁸, which integrates standardized clinical data from diabetologists across Japan, our study provides a broader and more representative analysis of real-world diabetic care.

Internationally, similar risk factors have been reported in diverse populations, including those in Europe¹⁴, the United States²⁸ and China^{13,29}. Previous landmark studies have primarily focused on long-term eGFR trajectories over extended periods: Krolewski *et al.*¹⁴ tracked patients for a median of 6.6 years to characterize linear decline patterns to ESRD, identifying higher HbA1c levels and elevated ACR as key risk factors for rapid progression. The CRIC study in the United States followed CKD patients for a median of 7.0 years and identified elevated plasma NT-proBNP as independent predictors of accelerated decline²⁸. Jiang *et al.*²⁹ followed patients for a median of 13.0 years to identify trajectory patterns of eGFR decline toward Stage 5 CKD/ESRD, with microalbuminuria and retinopathy as key predictors of accelerated eGFR decline. In another study, Jiang *et al.*¹³ derived a DKD risk prediction model from a systematic review and meta-analysis of 20 cohort studies and validated it in a Chinese cohort (median follow-up 2.9 years), identifying nine key predictors—age, BMI, smoking status, diabetic retinopathy, HbA1c, systolic blood pressure, HDL-C, triglycerides, and ACR—for early DKD. Chen *et al.*³⁰ conducted a 3-year prospective, multicenter cohort study and found that increased systolic blood pressure, DKD etiology, lower serum albumin, and more advanced CKD stage independently predicted a 50% eGFR decline or initiation of kidney replacement therapy. Although these studies vary considerably in design, population characteristics, and outcome definitions, several common risk factors emerge across different geographical regions and study settings. Our findings complement this body of evidence by demonstrating that short-term eGFR decline—rather than long-term slope—can be a practical and meaningful endpoint for risk stratification.

Low serum albumin may reflect underlying malnutrition or chronic inflammation. Previous studies have linked hyperglycemia-induced oxidative stress and inflammatory pathways to the pathogenesis of type 2 diabetes³¹. A large-scale UK cohort also demonstrated that reduced albumin levels were associated with increased incidence of diabetes and its microvascular complications¹², aligning with our observations.

Anemia, a frequent comorbidity in diabetic populations³², emerged as a robust risk factor for eGFR decline in our analysis, consistent with earlier reports^{7–10,16}. Prior studies have shown that hemoglobin levels are generally lower in diabetic individuals across CKD stages³³, and that anemia may manifest early in DKD progression^{33–35}. Moreover, anemia has been independently linked to both microvascular and macrovascular complications³⁶, and to accelerated renal function loss^{37,38}.

CHF was also identified as a shared risk factor for eGFR decline. The CRIC study²⁸ and other longitudinal analyses^{39,40} have demonstrated associations between cardiac biomarkers, proteinuria, and renal outcomes. Julián *et al.*⁴¹ conducted a prospective registry study of ambulatory patients with heart failure

Univariable

Multivariable

Table 4. (Continued)

Univariable	Multivariable				
	Total no. in category	No. of events (%)	OR [95% CI]	P-value	n = 3,205
Cerebrovascular disease	10,336	492 (4.76)	1.00		n = 4,355
Absent	821	55 (6.70)	1.44 [1.08–1.92]	0.014	
Present	10,793	505 (4.68)	1.00		
PAD	364	42 (11.54)	2.66 [1.90–3.71]	<0.001	
Absent	10,696	478 (4.47)	1.00		
Present	461	69 (14.97)	3.76 [2.87–4.94]	<0.001	
CHF					
Absent					
Present					

On the left side, the unadjusted odds ratio was calculated for each complication/comorbidity as the explanatory variable. Multivariable logistic regression analysis, employing each explanatory variable based on a significance level of $P \leq 0.1$, was performed to calculate the respective adjusted odds ratio, which is shown in the table on the middle side. Multiple logistic analysis was then performed again by selecting the variables with $P \leq 0.1$ in the middle analysis as explanatory variables, which are shown in the table on the right side. A, albumin: creatinine ratio; BMI, body mass index; CAD, coronary artery disease; CHF, chronic heart failure; CKD, chronic kidney disease; G, grade; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein cholesterol; PAD, peripheral artery disease; TG, triglycerides.

and type 2 diabetes and found that older age, nonischemic heart failure etiology, HFpEF or HFmrEF, poor glycemic control, and higher baseline eGFR were associated with more pronounced renal decline, and that decreased eGFR was independently associated with increased all-cause mortality. The concept of cardio-renal anemia syndrome—where heart failure, CKD, and anemia exacerbate one another—has gained attention^{22,23}, and our findings suggest this interplay may be particularly relevant in diabetic populations. Histopathological studies have further shown that interstitial fibrosis and tubular atrophy are linked to lower hemoglobin and worse renal outcomes⁴².

Hypertriglyceridemia, a common lipid abnormality in CKD⁴³, was also a consistent predictor. Prior research has shown that elevated TG and reduced HDL-C levels are associated with DKD development^{44,45}.

Some findings warrant further discussion. Although older age is generally considered a risk factor for DKD^{13,46}, our data suggest that younger individuals (<65 years) may be more susceptible to rapid eGFR decline over shorter intervals. Similarly, elevated HbA1c was predictive at 1 year but not at two, possibly due to reliance on baseline values rather than longitudinal glycemic trends. Conversely, smoking and hypertension were significant at 2 years only, implying that atherosclerotic mechanisms may exert greater influence over longer durations^{47,48}.

Several limitations should be acknowledged. First, medication data were not available, which may confound associations between clinical variables and renal outcomes⁴⁹. Second, patients could not be tracked across institutions, potentially limiting longitudinal follow-up. Third, iron status and erythropoietin levels—important modifiers of anemia-related renal risk⁵⁰—were not captured in the dataset. Moreover, the J-DREAMS registry is hospital-based and may not fully represent the general diabetic population, particularly those managed in primary care or community settings. Compared with community cohorts or claims databases, hospital registries may be less exhaustive in terms of population coverage. Therefore, extrapolation of our findings to broader populations should be done with caution.

However, the J-DREAMS database offers unique strengths that partially offset this limitation. It is specifically designed for diabetes care and includes detailed clinical information that is often unavailable in claims-based studies, such as laboratory test results, disease staging, and physician-confirmed diagnoses. These features allow for more precise phenotyping and risk stratification, which is critical for understanding renal prognosis in diabetes.

CONCLUSION

This study identified key clinical predictors of short-term eGFR decline in Japanese patients with diabetes using a large-scale real-world registry, demonstrating that beyond established contributors such as diabetic retinopathy and advanced kidney impairment, factors including anemia, reduced serum albumin,

and heart failure independently predict sustained eGFR decline in patients with diabetes.

By focusing on a clinically interpretable outcome and incorporating a broad patient population, our findings provide actionable insights for improving renal risk assessment and management in diabetic care.

ACKNOWLEDGMENTS

The authors extend their gratitude to the founding and current members of the J-DREAMS initiative at NCGM, including Drs. Takehiro Sugiyama, Hiroshi Kazio, Tetsuro Tsujimoto, and others. Appreciation is also given to Kiyori Komaki and Noriko Arimoto for their assistance with data management. This study was funded by Astellas Pharma Inc., Tokyo, Japan. J-DREAMS is also supported by grants from the Ministry of Health, Labour and Welfare Health Labour Sciences Research Grant (19FA1003 and 22FA1014 to K.U.) and Special Research (21CA2015 to K.U.) and the Japan Agency for Medical Research and Development Practical Research Project for Life-Style related Diseases including Cardiovascular Diseases and Diabetes Mellitus (15ek0210019 and 16ek0210019 to Hiroshi Kajio, 20ek0210134, 21ek0210134, 22ek0210134 to K.U.).

DISCLOSURE

K.Y. has no conflicts of interest to disclose regarding this article. M.O. reports honoraria for lectures from Astellas Pharma Inc., MSD K.K., Otsuka Pharmaceutical Co., Ltd., Kissei Pharmaceutical Co., Ltd., Teijin Pharma Ltd., Nippon Boehringer Ingelheim Co., Ltd., Sanofi K.K., Abbott Japan LLC, Sumitomo Dainippon Pharma Co., Ltd., Novo Nordisk Pharma Ltd., Kowa Co., Ltd., AstraZeneca K.K., Taisho Pharmaceutical Co., Ltd., Daiichi Sankyo Co., Ltd., Mitsubishi Tanabe Pharma Corporation, Ono Pharmaceutical Co., Ltd., Eli Lilly Japan K.K., Takeda Pharmaceutical Co., Ltd., Novartis Pharma K.K., and Bayer Yakuhin Ltd.; clinical commission/joint research grants from Novo nordisk Pharma Ltd., Eli Lilly Japan K.K., Nippon Boehringer Ingelheim Co., Ltd., Sumitomo Dainippon Pharma Co.m Ltd., Abbott Japan LLC, MSD K.K., Kyowa Kirin Co., Ltd., Astellas Pharma Inc., Sumitomo Dainippon Pharma Co., Ltd., and Bayer Yakuhin Ltd.; and scholarship grants from Sumitomo Dainippon Pharmac Co., Ltd. and Nippon Boehringer Ingelheim Co. K.U. reports honoraria for lectures from Sumitomo Pharma Co., Ltd., Novo Nordisk Pharma Ltd., Taisho Pharmaceutical Co., Ltd., Eli Lilly Japan K.K., MSD K.K., Abbott Japan LLC, Bayer Yakuhin, Ltd, Astellas Pharma Inc., Nippon Boehringer Ingelheim Co., Ltd., AstraZeneca K.K., ONO PHARMACEUTICAL Co., Ltd.; clinical commissioned/joint research grants from Sumitomo Dainippon Pharma Co., Ltd., Novo Nordisk Pharma Ltd., Eli Lilly Japan K.K., Abbott Japan LLC, and Sanofi K.K., Nippon Boehringer Ingelheim Co., Ltd., MSD K.K.; and scholarship grants from Nippon Boehringer Ingelheim Co., Ltd., Novo Nordisk Pharma Ltd., Sumitomo Dainippon Pharma Co., Ltd., Takeda Pharmaceutical Co., Ltd., Sanofi K.K. K.U. is an Editorial Board member of the Journal

of Diabetes Investigation and a co-author of this article. To minimize bias, they were excluded from all editorial decision-making related to the acceptance of this article for publication.

Approval of the research protocol: The study protocol was approved by the Institutional Review Board at the National Center for Global Health and Medicine (NCGM; approval number: NCGM-S-004378-00), and also by the Astellas Medical Affairs Japan Protocol Review Committee.

Informed consent: N/A.

Registry and the registration no. of the study/trial: N/A.

Animal studies: N/A.

DATA AVAILABILITY STATEMENT

Due to institutional and regulatory policies governing the J-DREAMS registry and the Japan Diabetes Society, individual-level data cannot be publicly shared. Researchers with appropriate credentials and study proposals may contact the JDS or NCGM for potential data access.

REFERENCES

- Magliano DJ, Boyko EJ, IDF Diabetes Atlas 10th Edition Scientific Committee. IDF Diabetes Atlas, 10th edn. Brussels: International Diabetes Federation, 2021.
- Ohsugi M, Eiki J, Iglay K, *et al.* Comorbidities and complications in Japanese patients with type 2 diabetes mellitus: retrospective analyses of J-DREAMS, an advanced electronic medical records database. *Diabetes Res Clin Pract* 2021; 178: 108845.
- Jha V, Garcia-Garcia G, Iseki K, *et al.* Chronic kidney disease: global dimension and perspectives. *Lancet* 2013; 382: 260–272.
- Tuttle KR, Bakris GL, Bilous RW, *et al.* Diabetic kidney disease: a report from an ADA consensus conference. *Diabetes Care* 2014; 37: 2864–2883.
- Ninomiya T, Perkovic V, de Galan BE, *et al.* Albuminuria and kidney function independently predict cardiovascular and renal outcomes in diabetes. *J Am Soc Nephrol* 2009; 20: 1813–1821.
- Ohkuma T, Jun M, Chalmers J, *et al.* Combination of changes in estimated GFR and albuminuria and the risk of major clinical outcomes. *Clin J Am Soc Nephrol* 2019; 14: 862–872.
- Ueda H, Ishimura E, Shoji T, *et al.* Factors affecting progression of renal failure in patients with type 2 diabetes. *Diabetes Care* 2003; 26: 1530–1534.
- Mohanram A, Zhang Z, Shahinfar S, *et al.* Anemia and end-stage renal disease in patients with type 2 diabetes and nephropathy. *Kidney Int* 2004; 66: 1131–1138.
- Babazono T, Hanai K, Suzuki K, *et al.* Lower haemoglobin level and subsequent decline in kidney function in type 2 diabetic adults without clinical albuminuria. *Diabetologia* 2006; 49: 1387–1393.
- Yamanouchi M, Furuichi K, Shimizu M, *et al.* Serum hemoglobin concentration and risk of renal function

- decline in early stages of diabetic kidney disease: a nationwide, biopsy-based cohort study. *Nephrol Dial Transplant* 2022; 37: 489–497.
11. Zhang J, Zhang R, Wang Y, *et al.* The level of serum albumin is associated with renal prognosis in patients with diabetic nephropathy. *J Diabetes Res* 2019; 2019: 7825804.
 12. Cai YW, Zhang HF, Gao JW, *et al.* Serum albumin and risk of incident diabetes and diabetic microvascular complications in the UK biobank cohort. *Diabetes Metab* 2023; 49: 101472.
 13. Jiang W, Wang J, Shen X, *et al.* Establishment and validation of a risk prediction model for early diabetic kidney disease based on a systematic review and meta-analysis of 20 cohorts. *Diabetes Care* 2020; 43: 925–933.
 14. Krolewski AS, Skupien J, Rossing P, *et al.* Fast renal decline to ESRD: An unrecognized feature of nephropathy in diabetes. *Kidney Int* 2017; 91: 1300–1311.
 15. Vistisen D, Andersen GS, Hulman A, *et al.* Progressive decline in estimated glomerular filtration rate in patients with diabetes after moderate loss in kidney function—even without albuminuria. *Diabetes Care* 2019; 42: 1886–1894.
 16. Hirano D, Unoki-Kubota H, Imasawa T, *et al.* Independent risk factors of rapid glomerular filtration rate decline in patients with type 2 diabetes with preserved kidney function and normoalbuminuria: a multicenter cohort study. *J Diabetes Investig* 2023; 14: 874–882.
 17. Yoshida Y, Kashiwabara K, Hirakawa Y, *et al.* Conditions, pathogenesis, and progression of diabetic kidney disease and early decliner in Japan. *BMJ Open Diabetes Res Care* 2020; 8: e000902.
 18. Sugiyama T, Miyo K, Tsujimoto T, *et al.* Design of and rationale for the Japan diabetes compREhensive database project based on an advanced electronic medical record system (J-DREAMS). *Diabetol Int* 2017; 8: 375–382.
 19. Seino Y, Nanjo K, Tajima N, *et al.* Report of the Committee on the Classification and Diagnostic Criteria of Diabetes Mellitus. *J Diabetes Investig* 2010; 1: 212–228.
 20. Tsubakihara Y, Nishi S, Akiba T, *et al.* Japanese Society for Dialysis Therapy: Guidelines for Renal Anemia in Chronic Kidney Disease. *Ther Apher Dial* 2010; 14: 240–275.
 21. Fioretto P, Caramori ML, Mauer M. The kidney in diabetes: dynamic pathways of injury and repair. The Camillo Golgi lecture 2007. *Diabetologia* 2008; 51: 1347–1355.
 22. Silverberg D. The cardio-renal anaemia syndrome: Does it exist? *Nephrol Dial Transplant* 2003; 18: viii7–viii12.
 23. McCullough PA. Anemia of cardiorenal syndrome. *Kidney Int Suppl* 2011; 11: 35–45.
 24. Itano S, Kanda E, Nagasu H, *et al.* eGFR slope as a surrogate endpoint for clinical study in early stage of chronic kidney disease: from the Japan chronic kidney disease database. *Clin Exp Nephrol* 2023; 27: 847–856.
 25. Odler B, Fu EL. eGFR slope as a primary endpoint for clinical trials of CKD progression: one size fits all? *Clin Kidney J* 2024; 17: sfae001.
 26. Inker LA, Heerspink HJL, Tighiouart H, *et al.* GFR slope as a surrogate end point for kidney disease progression in clinical trials: a meta-analysis of treatment effects of randomized controlled trials. *J Am Soc Nephrol* 2019; 30: 1735–1745.
 27. Vonesh E, Tighiouart H, Ying J, *et al.* Mixed-effects models for slope-based endpoints in clinical trials of chronic kidney disease. *Stat Med* 2019; 38: 4218–4239.
 28. Anderson AH, Xie D, Wang X, *et al.* Novel risk factors for progression of diabetic and nondiabetic CKD: Findings from the chronic renal insufficiency cohort (CRIC) study. *Am J Kidney Dis* 2021; 77: 56–73.e1.
 29. Jiang G, Luk AOY, Tam CHT, *et al.* Progression of diabetic kidney disease and trajectory of kidney function decline in Chinese patients with type 2 diabetes. *Kidney Int* 2019; 95: 178–187.
 30. Chen S, Chen L, Jiang H. Prognosis and risk factors of chronic kidney disease progression in patients with diabetic kidney disease and non-diabetic kidney disease: a prospective cohort CKD-ROUTE study. *Ren Fail* 2022; 44: 1309–1318.
 31. Oguntibeju OO. Type 2 diabetes mellitus, oxidative stress and inflammation: examining the links. *Int J Physiol Pathophysiol Pharmacol* 2019; 11: 45–63.
 32. Thomas MC, MacIsaac RJ, Tsalamandris C, *et al.* Unrecognized anemia in patients with diabetes. *Diabetes Care* 2003; 26: 1164–1169.
 33. Al-Khouri S, Afzali B, Shah N, *et al.* Anaemia in diabetic patients with chronic kidney disease—prevalence and predictors. *Diabetologia* 2006; 49: 1183–1189.
 34. El-Achkar TM, Ohmit SE, McCullough PA, *et al.* Higher prevalence of anemia with diabetes mellitus in moderate kidney insufficiency: the kidney early evaluation program. *Kidney Int* 2005; 67: 1483–1488.
 35. Alicic RZ, Tuttle KR. Management of the diabetic patient with advanced chronic kidney disease. *Semin Dial* 2010; 23: 140–147.
 36. Ito H, Takeuchi Y, Ishida H, *et al.* Mild anemia is frequent and associated with micro- and macroangiopathies in patients with type 2 diabetes mellitus. *J Diabetes Investig* 2010; 1: 273–278.
 37. Xie L, Shao X, Yu Y, *et al.* Anemia is a risk factor for rapid eGFR decline in type 2 diabetes. *Front Endocrinol* 2023; 14: 1052227.
 38. Meguro S, Tomita M, Kabeya Y, *et al.* Factors associated with the decline of kidney function differ among eGFR strata in subjects with type 2 diabetes mellitus. *Int J Endocrinol* 2012; 2012: 687867.
 39. Zhao Y, Zhao L, Wang Y, *et al.* The association of plasma NT-proBNP level and progression of diabetic kidney disease. *Ren Fail* 2023; 45: 2158102.
 40. Nohara N, Ito H, Matsumoto M, *et al.* Predictive factors associated with increased progression to dialysis in early chronic kidney disease (stage 1–3) patients. *Clin Exp Nephrol* 2016; 20: 740–747.

41. Julián MT, Codina P, Lupón J, *et al.* Long-term trajectory of estimated glomerular filtration rate in ambulatory patients with type 2 diabetes and heart failure: clinical insights and prognostic implications. *Cardiovasc Diabetol* 2025; 24: 104.
42. Shimizu M, Furuichi K, Kitajima S, *et al.* Impact of the relationship between hemoglobin levels and renal interstitial fibrosis on long-term outcomes in type 2 diabetes with biopsy-proven diabetic nephropathy. *BMC Nephrol* 2021; 22: 319.
43. Theofilis P, Vordoni A, Koukoulaki M, *et al.* Dyslipidemia in chronic kidney disease: contemporary concepts and future therapeutic perspectives. *Am J Nephrol* 2021; 52: 693–701.
44. Russo GT, De Cosmo S, Viazzi F, *et al.* Plasma triglycerides and HDL-C levels predict the development of diabetic kidney disease in subjects with type 2 diabetes: the AMD annals initiative. *Diabetes Care* 2016; 39: 2278–2287.
45. Lamprea-Montealegre JA, Staplin N, Herrington WG, *et al.* Apolipoprotein B, triglyceride-rich lipoproteins, and risk of cardiovascular events in persons with CKD. *Clin J Am Soc Nephrol* 2020; 15: 47–60.
46. Moriya T, Tanaka S, Sone H, *et al.* Patients with type 2 diabetes having higher glomerular filtration rate showed rapid renal function decline followed by impaired glomerular filtration rate: Japan diabetes complications study. *J Diabetes Complicat* 2017; 31: 473–478.
47. Sawicki PT, Didjurgeit U, Mühlhauser I, *et al.* Smoking is associated with progression of diabetic nephropathy. *Diabetes Care* 1994; 17: 126–131.
48. Ito H, Antoku S, Furusho M, *et al.* The prevalence of the risk factors for atherosclerosis among type 2 diabetic patients is greater in the progressive stages of chronic kidney disease. *Nephron Extra* 2013; 3: 66–72.
49. Neuen BL, Heerspink HJL, Vart P, *et al.* Estimated lifetime cardiovascular, kidney, and mortality benefits of combination treatment with SGLT2 inhibitors, GLP-1 receptor agonists, and nonsteroidal MRA compared with conventional care in patients with type 2 diabetes and albuminuria. *Circulation* 2024; 149: 450–462.
50. Fujita Y, Doi Y, Hamano T, *et al.* Low erythropoietin levels predict faster renal function decline in diabetic patients with anemia: a prospective cohort study. *Sci Rep* 2019; 9: 14871.